

AMAZON | BOTTLENECK TO ROBUST QUALITY

AN ADVANCED QUALITY ENGINEERING FRAMEWORK USING THEORY OF CONSTRAINTS, TAGUCHI ROBUST DESIGN AND DOE

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SECTION 01: CASE OBJECTIVE

The objective of this case study is to explore how Quality Engineering principles can be used to improve performance in a complex large scale process by first identifying the system constraint and then designing an improvement that remains robust under variation.

The framework combines Theory of Constraints with Taguchi Robust Design and Design of Experiments. FMEA is then used to evaluate implementation risks, followed by controlled validation and continuous improvement through PDCA and Kaizen.

The central idea is that improving quality should not focus only on making one activity faster or more efficient. The improvement should strengthen the overall system while remaining stable when operating conditions change.

SECTION 02: QUALITY PROBLEM

In a large scale digital and operational environment, performance can be affected by multiple interconnected stages. A delay or limitation in one stage can influence the performance of the entire system.

Improving every stage simultaneously may consume resources without addressing the factor that is actually limiting overall performance.

The Quality Engineering challenge is therefore to understand the complete process, identify the most critical constraint, analyse the conditions influencing that constraint and develop a robust improvement that can perform consistently across relevant variation.

SECTION 03: PROCESS ASSESSMENT & CONSTRAINT IDENTIFICATION

The first step is to understand the complete process rather than immediately modifying individual activities. A process level assessment is used to observe the flow, performance variation, resource utilisation, operating conditions and potential limitations across the system.

Theory of Constraints is then applied to identify the critical bottleneck that is restricting overall system performance. The improvement priority is based on system constraint rather than simply selecting the stage that appears easiest to improve.

PROCESS ASSESSMENT → SYSTEM ANALYSIS →
CONSTRAINT IDENTIFICATION → CONSTRAINT
IMPROVEMENT

SECTION 04: HISTORICAL DATA & OPERATING CONDITION ANALYSIS

Once the critical constraint is identified, historical performance data and relevant operating conditions are analysed. The objective is to understand when the constraint becomes significant, what factors influence its performance and whether the observed limitation remains consistent across different conditions.

This analysis creates the foundation for experimental investigation rather than relying on assumptions or isolated observations. Relevant factors should be identified before experimentation so that the subsequent DOE is based on meaningful process knowledge.

SECTION 05: FROM CONSTRAINT TO EXPERIMENTATION

After the critical constraint and relevant influencing factors are understood, Taguchi Robust Design and Design of Experiments can be used to systematically investigate the relationship between process factors and performance.

Instead of changing factors randomly, controlled experimentation allows multiple conditions to be evaluated in a structured manner. The objective is not simply to find the best result under one ideal condition. The objective is to identify a combination that remains stable when relevant operating variation and noise factors are present.

CONSTRAINT → FACTORS → DOE → ROBUST DESIGN →
VALIDATION

CORE QUALITY CONCEPT

Theory of Constraints × Taguchi Robust Design × Design of Experiments × FMEA × PDCA × Kaizen

This integrated methodology ensures that operational bottlenecks are systematically isolated, tested, controlled, and permanently standardized into the operational workflow.

AMAZON | FRAMEWORK CONTINUATION & VALIDATION

ADVANCED RISK MANAGEMENT, CONTROLLED PILOT, AND CONTINUOUS IMPROVEMENT CYCLE

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SECTION 06: TAGUCHI ROBUST DESIGN & DOE

Taguchi Robust Design is used to investigate how process factors influence performance while considering variation. Design of Experiments provides a structured method for testing combinations of relevant factors rather than relying on one factor at a time experimentation.

The objective is to identify operating conditions that provide consistent performance and reduce sensitivity to variation. The result should be evaluated not only by average performance but also by stability and consistency under different relevant conditions.

SECTION 07: ROBUSTNESS UNDER VARIATION

An improvement that performs well only under ideal conditions may not provide reliable system performance. The proposed framework therefore considers operating variation and noise factors during experimentation and validation.

The preferred solution is the one that provides reliable performance across relevant conditions rather than producing a strong result under only one controlled condition. This converts improvement from simple optimisation into robust Quality Engineering.

SECTION 08: FMEA & IMPLEMENTATION RISK

Before wider implementation, potential failure modes associated with the proposed improvement are evaluated through FMEA. Severity, Occurrence and Detection are considered to prioritise risks and determine where preventive action is required.

FMEA is not treated as a replacement for engineering judgement. High severity risks should receive appropriate attention even when numerical prioritisation alone does not fully represent the risk. Preventive controls are then defined before the solution moves toward broader implementation.

**FAILURE MODE → SEVERITY → OCCURRENCE →
DETECTION → PREVENTIVE CONTROL**

SECTION 09: CONTROLLED PILOT & VALIDATION

The validated solution should not immediately be introduced across the complete system. A controlled pilot is first conducted under defined conditions. Performance is then measured across different relevant operating conditions and compared with the established baseline.

**CONTROLLED PILOT → VARIATION VALIDATION →
RESTRUCTURE & RETEST**

SECTION 10: TOC AS A CONTINUOUS CYCLE

Theory of Constraints does not end after the first bottleneck is improved. Once the identified constraint is improved, the overall system should be reassessed. The improvement may cause another stage to become the new limiting factor.

The Quality Engineering cycle therefore continues by identifying the next constraint and repeating the improvement process. This prevents the organisation from optimising one area while ignoring the next system limitation.

**IDENTIFY → IMPROVE → REASSESS → NEXT
CONSTRAINT → CONTINUE**

SECTION 11: PDCA, KAIZEN & STANDARDISATION

After successful validation, the improved process should be standardised so that the validated method becomes part of normal operation. PDCA provides the continuous improvement structure:

- **Plan:** Defines problem, baseline and approach.
- **Do:** Implements validated solution under control.
- **Check:** Evaluates performance and variation.
- **Act:** Standardises improvement or restructures.

Kaizen keeps the process open to small continuous improvements rather than treating one successful implementation as the final state of quality.

SECTION 12: FINAL QUALITY FRAMEWORK

**PROCESS → TOC → HISTORICAL DATA → TAGUCHI → DOE →
ROBUSTNESS → FMEA → PILOT → VALIDATION → PDCA →
KAIZEN → STANDARDISATION**

Key Quality Insight: Improving the constraint is not the end of quality improvement. It is the beginning of the next improvement cycle. A strong Quality Engineering system identifies what limits performance, investigates influencing factors, designs robust solutions, evaluates risk, validates under variation and continuously improves through evidence.

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